12TH BIOLOGY · CHAPTER 25 · FULL CHAPTER MCQS

Ecology

75 topic-wise MCQs from the complete chapter with answers and reasons

25.1 LEVELS OF ECOLOGICAL ORGANIZATION

1. The study of the relationships of organisms with each other and the environment is:

(a) Ecology (b) Evolution
(c) Genetics (d) Taxonomy

Answer: (a) Ecology — Definition of ecology.

2. The basic unit of biological classification is the:

(a) Population (b) Community
(c) Species (d) Biome

Answer: (c) Species — Its members interbreed to produce fertile offspring.

3. All the Markhors of Chitral Gol National Park form a:

(a) Species (b) Biome
(c) Community (d) Population

Answer: (d) Population — Same species, same area, same time.

4. All the different populations living and interacting in an area form a:

(a) Population (b) Community
(c) Species (d) Biosphere

Answer: (b) Community — It includes plants, animals, fungi and bacteria.

5. An ecosystem consists of:

(a) Biotic components only (b) Abiotic components only
(c) Biotic and abiotic components (d) One species

Answer: (c) Biotic and abiotic components — Community plus the non-living environment.

6. The Thar Desert is an example of a/an:

(a) Population (b) Biosphere
(c) Species (d) Ecosystem

Answer: (d) Ecosystem — It has a community and abiotic factors.

7. The coniferous forest biome of Pakistan is located in:

(a) Murree and Swat (b) Thal and Thar
(c) Karachi (d) Badin

Answer: (a) Murree and Swat — High-altitude areas with cold winters.

studysapiens.com

8. The highest level of ecological organization is the:

(a) Biome (b) Biosphere
(c) Ecosystem (d) Community

Answer: (b) Biosphere — The global sum of all biomes.

9. The biosphere is often called the:

(a) Global ecosystem (b) Global village
(c) Largest biome (d) Lithosphere

Answer: (a) Global ecosystem — Land, water and air interact in it.

25.2 BIOGEOCHEMICAL CYCLES

10. A primary reservoir is also called a:

(a) Source (b) Sink or pool
(c) Niche (d) Habitat

Answer: (b) Sink or pool — The element is stored there for long periods.

11. In sedimentary cycles, the reservoir is the:

(a) Air (b) Living body
(c) Ocean (d) Earth's crust

Answer: (d) Earth's crust — For example the phosphorus cycle.

12. Which of the following is a sedimentary cycle?

(a) Phosphorus cycle (b) Nitrogen cycle
(c) Carbon cycle (d) Water cycle

Answer: (a) Phosphorus cycle — Also sulfur and calcium cycles.

13. The water cycle is also called the:

(a) Geological cycle (b) Sedimentary cycle
(c) Hydrological cycle (d) Trophic cycle

Answer: (c) Hydrological cycle — Continuous movement of water.

14. The entire water cycle is powered by:

(a) Wind (b) The sun
(c) Gravity (d) Plants

Answer: (b) The sun — Solar radiation causes evaporation.

15. About how much of global evaporation occurs from the oceans?

(a) 50% (b) 78%
(c) 10% (d) 86%

Answer: (d) 86% — As given in the textbook.

studysapiens.com

16. Rain mingled with snow or hail is called:

(a) Sleet (b) Dew
(c) Fog (d) Frost

Answer: (a) Sleet — A form of precipitation.

17. Seeping of water into the soil is called:

(a) Runoff (b) Condensation
(c) Infiltration (d) Transpiration

Answer: (c) Infiltration — It reaches the water table.

18. The major exit point of water from plants is:

(a) Respiration (b) Photosynthesis
(c) Guttation (d) Transpiration

Answer: (d) Transpiration — Through the stomata of leaves.

19. The percentage of nitrogen gas in the atmosphere is:

(a) 78% (b) 21%
(c) 86% (d) 0.04%

Answer: (a) 78% — N_2 has a strong triple bond.

20. More than 90% of nitrogen fixation is:

(a) Atmospheric (b) Biological
(c) Industrial (d) Volcanic

Answer: (b) Biological — By soil microorganisms.

21. A symbiotic nitrogen-fixing bacterium of legume root nodules is:

(a) Azotobacter (b) Clostridium
(c) Rhizobium (d) Nitrobacter

Answer: (c) Rhizobium — In chickpea and mung bean.

22. Atmospheric fixation of nitrogen is brought about by:

(a) Lightning (b) Bacteria
(c) Fertilizers (d) Fungi

Answer: (a) Lightning — It fixes only 5–10%.

23. Nitrobacter convert:

(a) Ammonia to nitrite (b) Nitrate to N_2
(c) Nitrite to nitrate (d) N_2 to ammonia

Answer: (c) Nitrite to nitrate — Second step of nitrification.

24. The use of nitrates by plants to make proteins and DNA is:

(a) Ammonification (b) Assimilation
(c) Nitrification (d) Denitrification

Answer: (b) Assimilation — Entry into the biotic world.

studysapiens.com

25. Conversion of organic nitrogen back into ammonia by decomposers is:

(a) Nitrification (b) Fixation
(c) Assimilation (d) Ammonification

Answer: (d) Ammonification — The recycling step.

26. Denitrification occurs mainly in:

(a) Dry sandy soils (b) Waterlogged soils
(c) The atmosphere (d) Root nodules

Answer: (b) Waterlogged soils — By *Pseudomonas* and *Thiobacillus*.

25.3 THE FLOW OF ENERGY

27. Trophic level 1 consists of:

(a) Primary producers (b) Herbivores
(c) Carnivores (d) Decomposers

Answer: (a) Primary producers — Green plants, algae, photosynthetic bacteria.

28. Herbivores belong to trophic level:

(a) 1 (b) 3
(c) 2 (d) 4

Answer: (c) 2 — They are primary consumers.

29. A frog or snake feeding on herbivores is a:

(a) Producer (b) Secondary consumer
(c) Primary consumer (d) Detritivore

Answer: (b) Secondary consumer — Trophic level 3.

30. Wolf and hawk are examples of:

(a) Primary consumers (b) Decomposers
(c) Producers (d) Tertiary consumers

Answer: (d) Tertiary consumers — They eat carnivores.

31. Dead, decomposing plant and animal material is called:

(a) Detritus (b) Humus only
(c) Biomass (d) Sediment

Answer: (a) Detritus — Detritivores feed on it.

32. How much solar energy falling on producers is converted to organic matter?

(a) 10% (b) 90%
(c) 50% (d) 1 to 2%

Answer: (d) 1 to 2% — As bond-energy.

studysapiens.com

33. If producers fix 10,000 kcal, secondary consumers get about:

(a) 1,000 kcal (b) 10 kcal
(c) 100 kcal (d) 5,000 kcal

Answer: (c) 100 kcal — 10% of 10% of 10,000.

34. Energy during transfer between trophic levels is lost as:

(a) Light (b) Heat
(c) Sound (d) Matter

Answer: (b) Heat — So short chains are more efficient.

25.4 CHARACTERISTICS OF POPULATION

35. The statistical study of the characteristics of populations is:

(a) Ecology (b) Demography
(c) Biogeography (d) Taxonomy

Answer: (b) Demography — Population size is its key feature.

36. Humans and mammals show which survivorship curve?

(a) Type II (b) Type III
(c) Type I (d) Type IV

Answer: (c) Type I — Convex curve.

37. A constant probability of death throughout life gives a curve that is:

(a) Convex (b) S-shaped
(c) Concave (d) Diagonal

Answer: (d) Diagonal — Type II: birds, lizards, rodents.

38. Very high juvenile mortality is seen in:

(a) Fishes (b) Humans
(c) Birds (d) Rodents

Answer: (a) Fishes — Type III (concave).

39. In the exponential model, the population grows:

(a) Up to a limit (b) Not at all
(c) Negatively (d) Without limits

Answer: (d) Without limits — At its maximal rate.

40. The shape of the logistic growth curve is:

(a) J-shaped (b) Sigmoid (S-shaped)
(c) Straight line (d) Bell-shaped

Answer: (b) Sigmoid (S-shaped) — It flattens at the carrying capacity.

studysapiens.com

41. The size at which a logistic population stabilizes is its:

(a) Density (b) Viable size
(c) Carrying capacity (d) Dispersion

Answer: (c) Carrying capacity — Set by limited resources.

42. The number of individuals per unit area or volume is population:

(a) Density (b) Growth
(c) Dispersion (d) Survivorship

Answer: (a) Density — For example trees per square kilometre.

43. Nesting penguins show which spacing?

(a) Random (b) Uniform
(c) Clumped (d) None

Answer: (b) Uniform — They maintain defined territories.

44. The most common pattern of spacing in nature is:

(a) Random (b) Linear
(c) Uniform (d) Clumped

Answer: (d) Clumped — Resources are unevenly distributed.

45. Wind-dispersed seeds germinating wherever they fall give:

(a) Uniform spacing (b) Clumped spacing
(c) Random spacing (d) No spacing

Answer: (c) Random spacing — It is not common in nature.

46. A population below its minimal viable size faces a high risk of:

(a) Extinction (b) Overgrowth
(c) Migration (d) Mutation

Answer: (a) Extinction — Due to inbreeding and disasters.

47. The world population reached 8 billion in:

(a) 2011 (b) 2037
(c) 2022 (d) 2050

Answer: (c) 2022 — The Day of Eight Billion.

48. Pakistan's population in the 2023 census was:

(a) 207.7 million (b) 241.49 million
(c) 250 million (d) 180 million

Answer: (b) 241.49 million — Growth rate 2.55% annually.

49. In world population, Pakistan ranks:

(a) 3rd (b) 10th
(c) 7th (d) 5th

Answer: (d) 5th — The 5th most populous country.

studysapiens.com

50. Plants make about what share of the Earth's biomass?

(a) 82% (b) 13%
(c) 0.01% (d) 50%

Answer: (a) 82% — Bacteria make about 13%.

25.5 GREENHOUSE EFFECT

51. The most significant greenhouse gas is:

(a) Methane (b) CFCs
(c) Nitrous oxide (d) Carbon dioxide

Answer: (d) Carbon dioxide — It contributes about 80% of global warming.

52. Methane contributes about how much of global warming?

(a) 80% (b) 11%
(c) 6% (d) 3%

Answer: (b) 11% — From coal and gas production and livestock.

53. Enteric fermentation in cattle produces:

(a) Methane (b) CO_2
(c) Nitrous oxide (d) Ozone

Answer: (a) Methane — A livestock source.

54. Greenhouse gases emitted from old refrigerators are:

(a) Methane (b) Nitrous oxide
(c) Fluorinated gases (d) Carbon dioxide

Answer: (c) Fluorinated gases — Synthetic gases, about 3%.

55. According to the IPCC (2023), global surface temperature has risen by about:

(a) 1.1°C (b) 0.1°C
(c) 5°C (d) 11°C

Answer: (a) 1.1°C — Compared with 1850–1900.

56. The pre-industrial CO_2 concentration was about:

(a) 428 ppm (b) 100 ppm
(c) 280 ppm (d) 780 ppm

Answer: (c) 280 ppm — It is 428 ppm in 2026.

57. Jacobabad and Sibi are mentioned as examples of:

(a) Floods (b) Sea level rise
(c) Glacial melting (d) Heatwaves

Answer: (d) Heatwaves — Among the hottest places on Earth.

studysapiens.com

58. GLOF stands for:

(a) Global Level of Floods
(b) Glacial Lake Outburst Flood
(c) Greenhouse Layer of Fluorine
(d) Glacier Loss of Flow

Answer: (b) Glacial Lake Outburst Flood — Sudden release of water from a glacial lake.

59. Seawater intrusion threatens the coastal areas of:

(a) Murree and Swat
(b) Mianwali
(c) Chitral
(d) Thatta and Badin

Answer: (d) Thatta and Badin — Fertile land turns into salty marshes.

60. Turning of fertile land into desert is called:

(a) Deforestation
(b) Erosion
(c) Desertification
(d) Salinity

Answer: (c) Desertification — Heat evaporates soil moisture.

61. Pakistan's share of global GHG emissions is:

(a) About 10%
(b) Less than 1%
(c) About 5%
(d) Over 20%

Answer: (b) Less than 1% — Yet it is highly vulnerable.

25.6 IMPORTANT ECOSYSTEMS OF PAKISTAN

62. Temperate deciduous forests are found in:

(a) Thal and Thar
(b) Keti Bandar
(c) Shogran, Neelam Valley and Murree
(d) Bahawalpur

Answer: (c) Shogran, Neelam Valley and Murree — Moist mountain regions.

63. Annual rainfall in the temperate deciduous forest is:

(a) 750–1500 mm
(b) 200–750 mm
(c) Below 100 mm
(d) Above 3000 mm

Answer: (a) 750–1500 mm — Temperature 4°C to 30°C.

64. *Cedrus deodara* is commonly called:

(a) Blue Pine
(b) Himalayan Cedar
(c) Yew
(d) Kikar

Answer: (b) Himalayan Cedar — Flora of the temperate forest.

65. The most widespread terrestrial ecosystem of the plains of Pakistan is the:

(a) Mangrove forest
(b) Temperate forest
(c) Alpine meadow
(d) Tropical thorn forest

Answer: (d) Tropical thorn forest — It covers the Indus Plains.

66. Plants of the tropical thorn forest are:

(a) Hydrophytic (b) Halophytic
(c) Xerophytic (d) Epiphytic

Answer: (c) Xerophytic — Drought-resistant, with thorns.

67. *Prosopis cineraria* is locally called:

(a) Jhand (b) Karir
(c) Kikar (d) Peelu

Answer: (a) Jhand — Also called Kandi.

68. The houbara bustard and chinkara are fauna of the:

(a) Mangrove forest (b) Tropical thorn forest
(c) Alpine meadows (d) Temperate forest

Answer: (b) Tropical thorn forest — With camels and golden jackals.

69. High-altitude "cold deserts" are the:

(a) Mangroves (b) Deciduous forests
(c) Thorn forests (d) Alpine and sub-alpine meadows

Answer: (d) Alpine and sub-alpine meadows — Too harsh for large trees.

70. The Deosai Plains are an example of:

(a) Thorn forest (b) Mangrove forest
(c) Alpine meadows (d) Desert biome

Answer: (c) Alpine meadows — Mean annual temperature 1°C to 10°C.

71. The snow leopard and Himalayan ibex live in the:

(a) Indus delta (b) Thorn forest
(c) Thar Desert (d) Alpine meadows

Answer: (d) Alpine meadows — With marmots and musk deer.

72. Mangroves are often called:

(a) The Forests of the Sea (b) Cold deserts
(c) Natural jungles (d) Rain forests

Answer: (a) The Forests of the Sea — They grow where land meets sea.

73. *Avicennia marina* makes up what share of the mangrove forest?

(a) 50% (b) 95%
(c) 10% (d) 75%

Answer: (b) 95% — Locally called Timmar.

74. The water of the mangrove ecosystem is:

(a) Brackish (b) Fresh
(c) Distilled (d) Acidic

Answer: (a) Brackish — A mix of river water and seawater.

studysapiens.com

75. Mangrove trees have “breathing roots” because the soil is:

(a) Dry (b) Sandy

(c) Anaerobic (d) Frozen

Answer: (c) Anaerobic — Oxygen-poor mudflats.

studysapiens.com